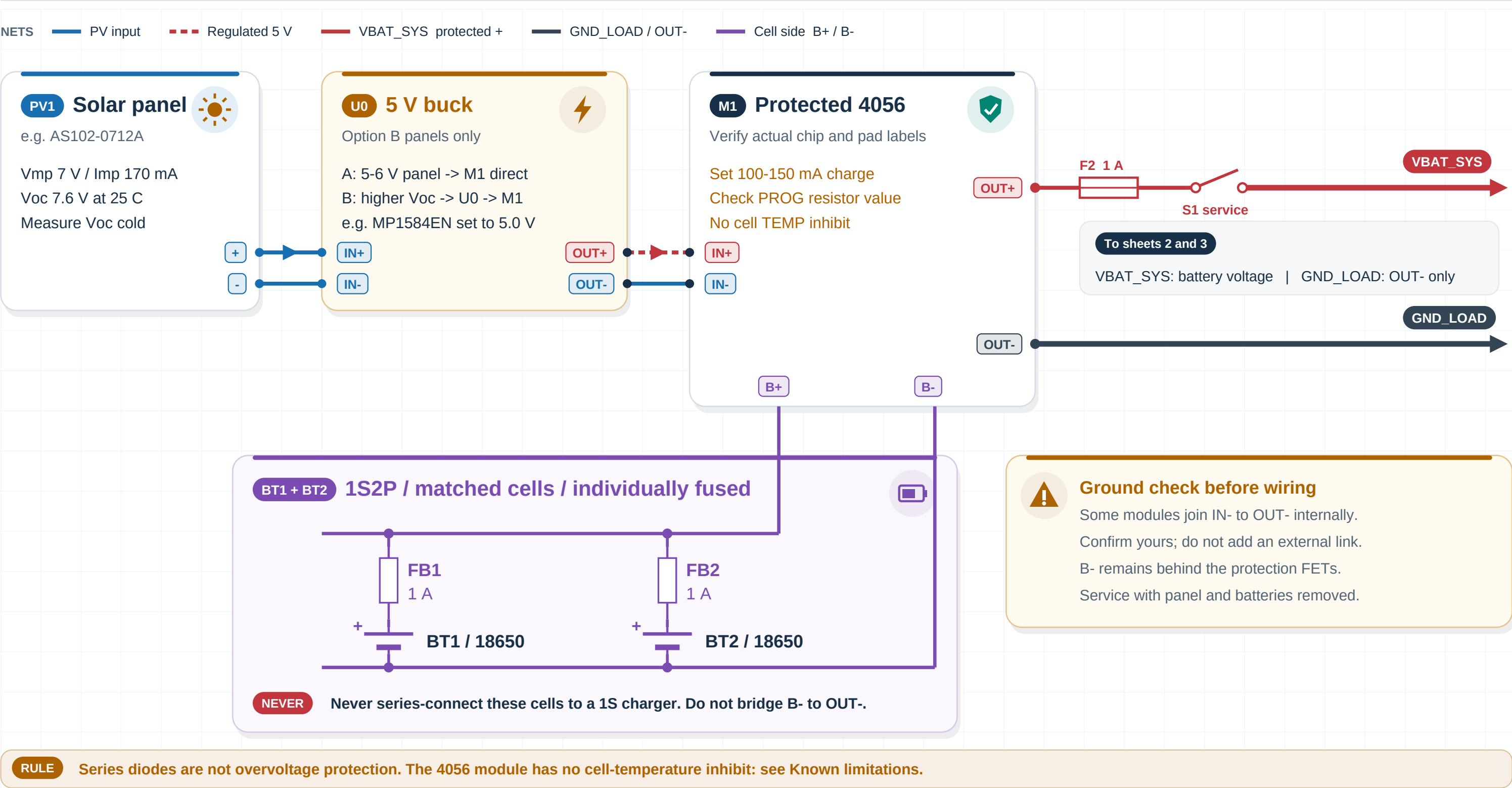


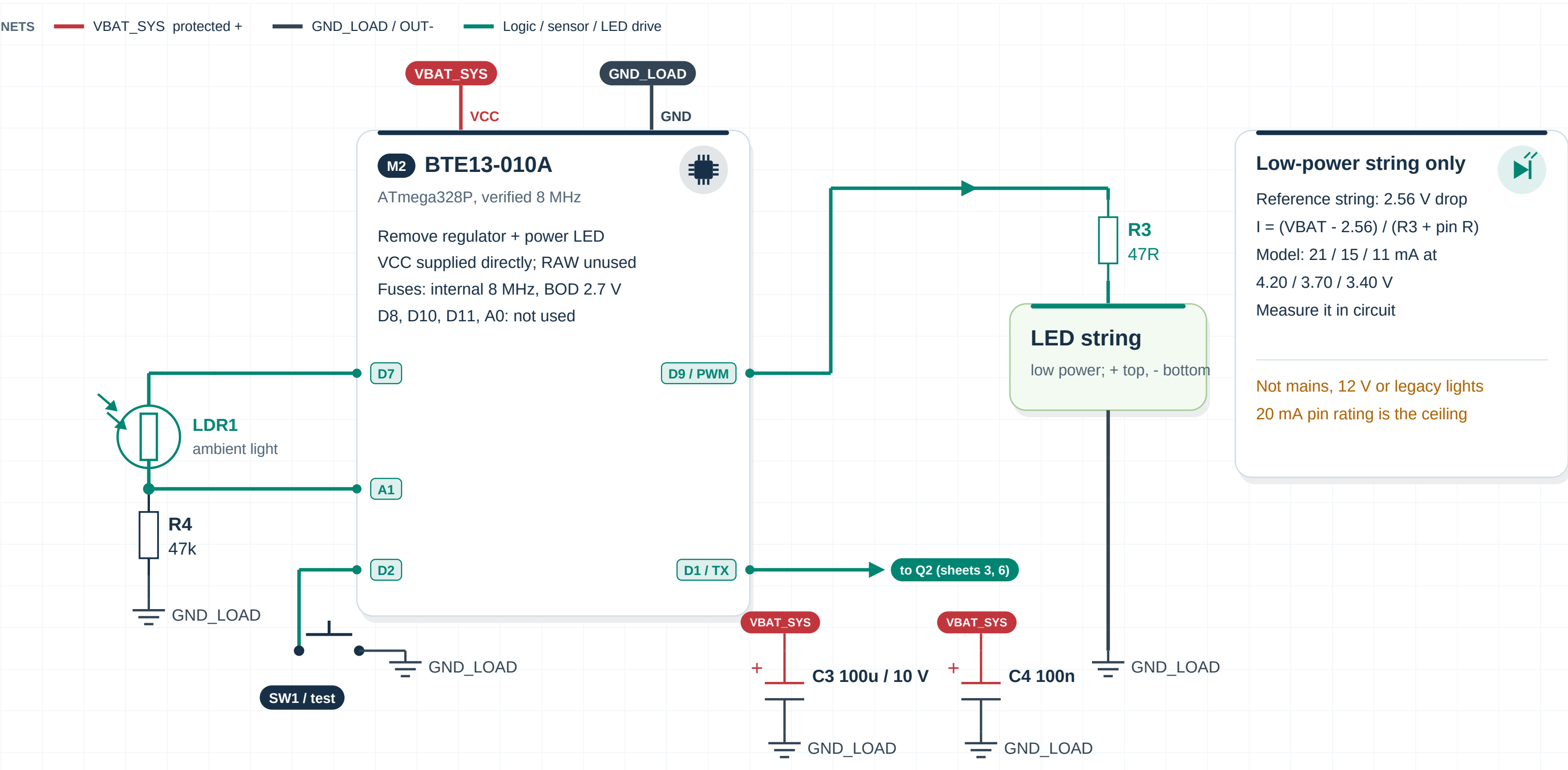
Module A: power, charging & battery

Terminal-level schematic. Amber blocks depend on your panel choice. All loads connect to the protected outputs.



Modules B-D: controller, light sensor & low-power LEDs

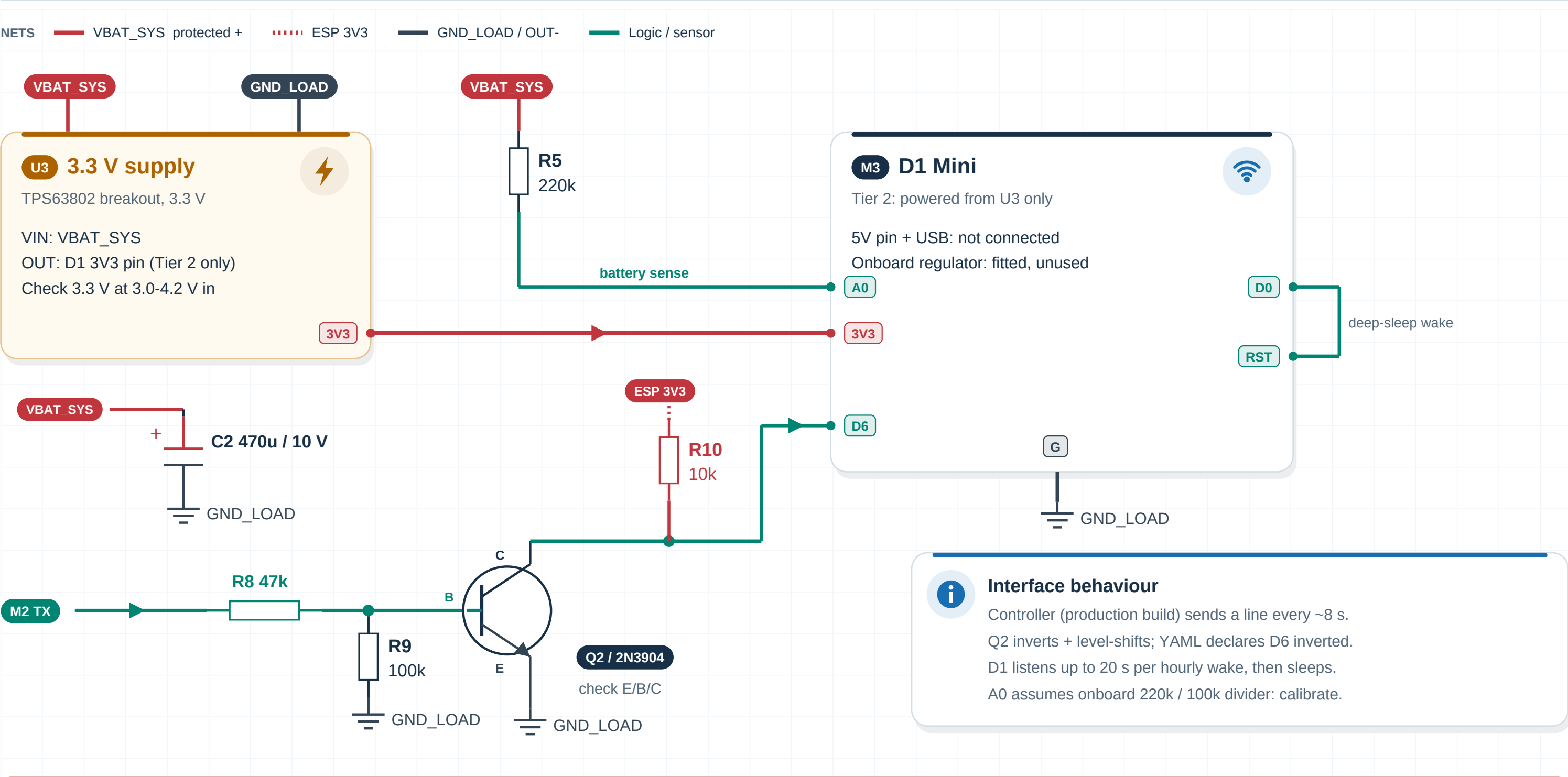
ATmega328P Pro Mini on its internal 8 MHz clock. D9 drives the low-power LED string directly by PWM through R3; no driver transistor.



RULE All ground symbols on this sheet mean OUT-. R3 sets the peak current; duty changes the average only. Verify current at 4.20 V and 100% duty.

Module E: telemetry, Tier 2 production (D1 Mini)

Hourly deep sleep on battery power. Same Q2 serial harness as Tier 1 (sheet 6); the D1 Mini is powered from U3.



SAFETY Disconnect external power AND signal leads before USB/programmer servicing. Never supply the ESP 3V3 pin from the cell directly.

WEATHERPROOF BOX / SHADED POSITION

PANEL **LIGHTS** **LDR**

PERFBOARD / MODULE CARRIER

U0 Input 5 V
Option B panels only

M1 Charger
IN / B / OUT labelled

M2 Pro Mini
8 MHz; sockets; ISP access

LED output
R3 only - D9 PWM direct

U3 3V3
TPS63802

M3 D1
Sockets / service

ANTENNA

Protected rails

BT1 / 18650 **BT2 / 18650**

Insulated parallel holder; fuse each + lead; separate cells from heat.

Build in this order

- 1 Prepare, fuse and flash the Pro Mini.
- 2 Add R3 (47R) + low-power LED string.
- 3 Add LDR + button; check dusk/dawn, LVC.
- 4 Qualify charger (and U0 if fitted).
- 5 Assemble the protected, fused pack.
- 6 Add telemetry; commission with Tier 1.

Label every connector

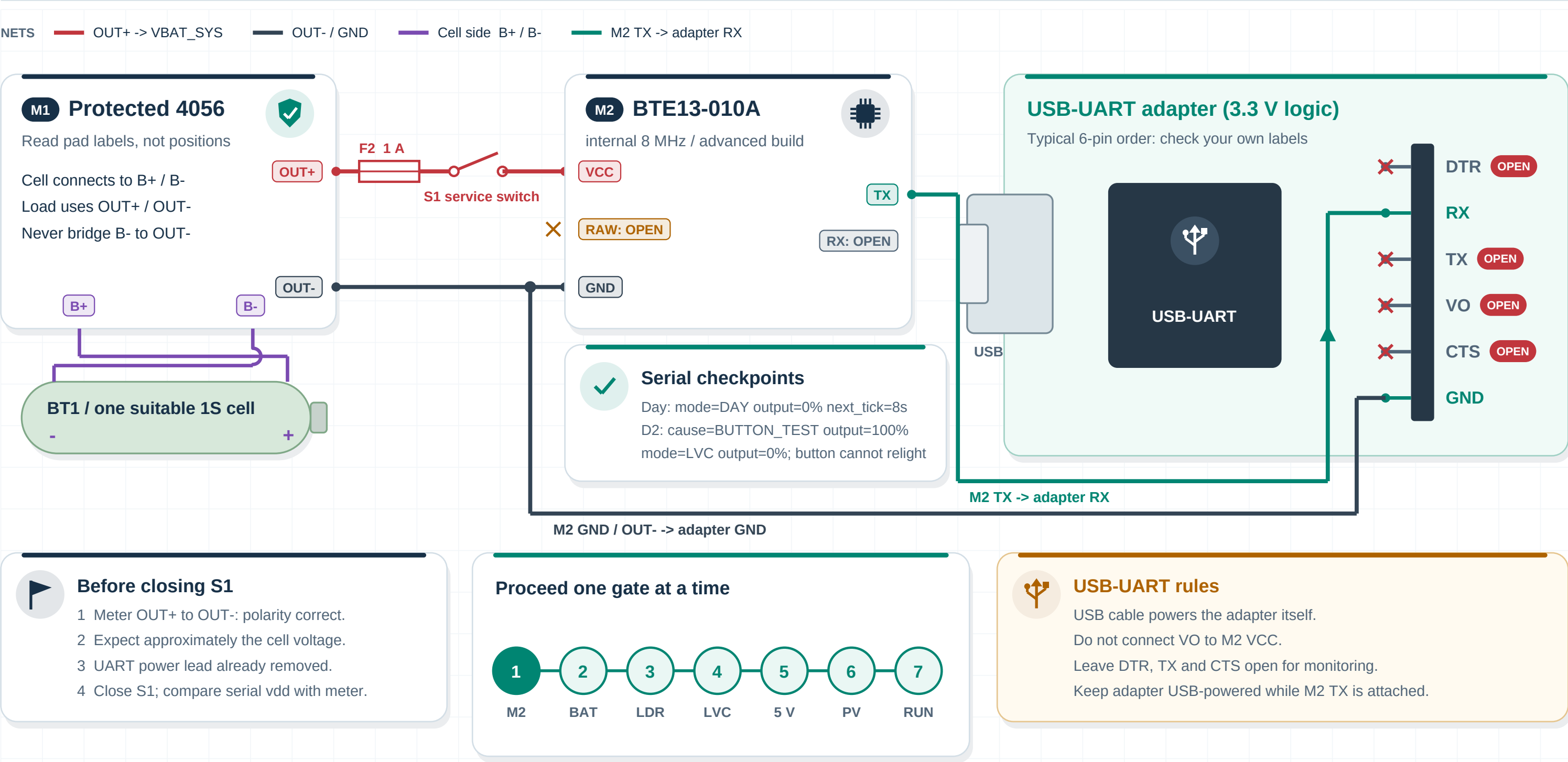
- RED: protected positive / VBAT_SYS
- BLACK: load return / OUT-
- BLUE: panel input pair
- PURPLE: battery-only connections
- GREEN: logic / sensor signals

Before closing the lid

- ☐ Strain relief, polarity, cell insulation.
- ☐ No exposed underside copper.
- ☐ Charger/cell temperature tests passed.
- ☐ 14-day Tier 1 soak test passed.

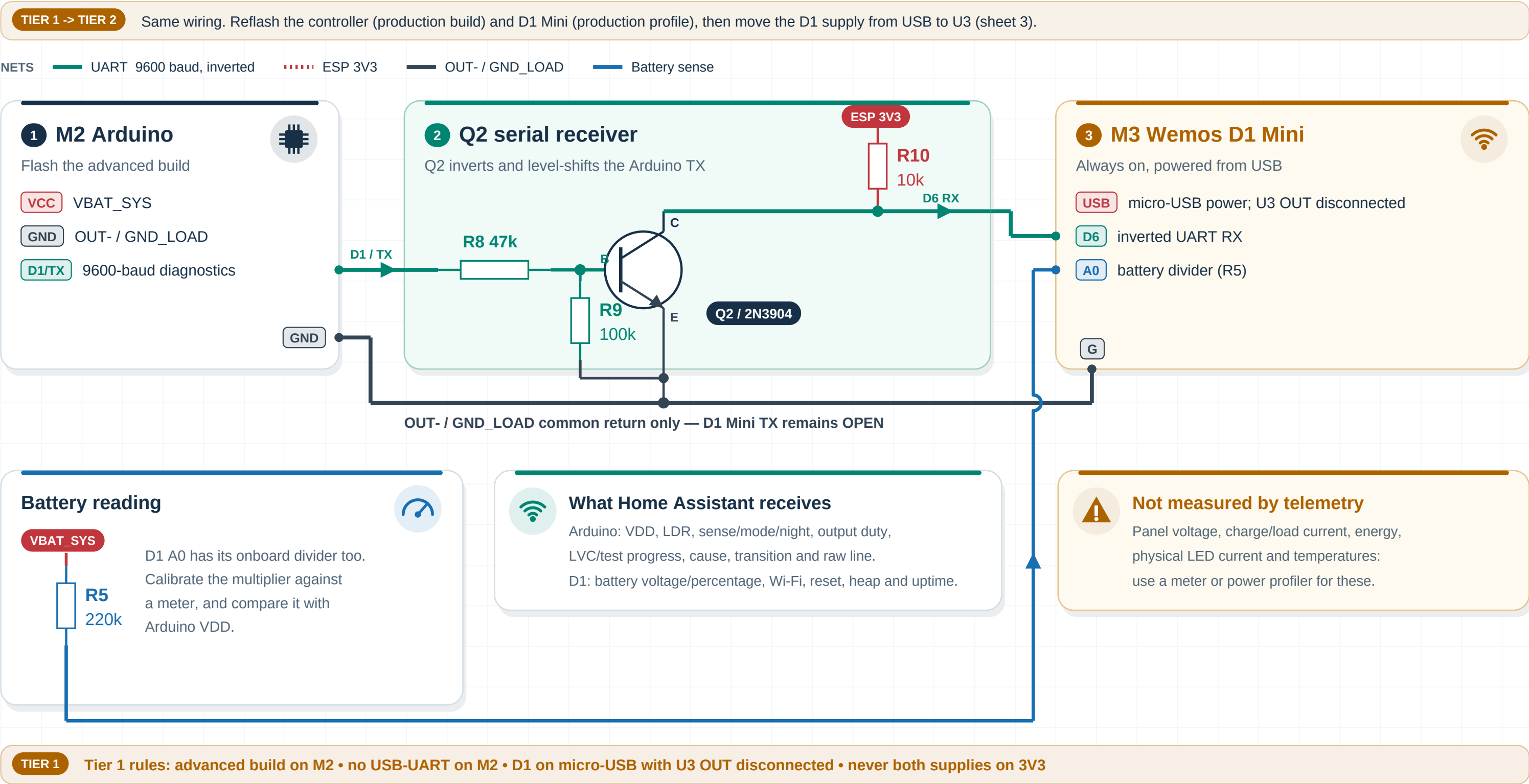
Staged power-up & wired serial monitor

Optional alternative to Tier 1 telemetry. The USB-UART adapter only listens: connect RX and GND; leave its power and control pins open.



Module E: telemetry, Tier 1 Advanced (D1 Mini)

Always-on commissioning instrument: decodes every controller diagnostic line into Home Assistant while you build, calibrate and soak-test.



System overview

Every module on one sheet. Telemetry reports controller state to Home Assistant; it is not a current or energy meter.

